

Earthquake Risk Analysis with Extreme Value Theory

POT-GPD Tail Modelling and Spatial Forecasting

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Contents

1	Introduction	1
2	Dataset and Preprocessing	1
2.1	Source and Scope	1
2.2	Selected Variables	1
2.3	Cleaning Procedure	1
2.4	Descriptive Overview	2
3	EVT Modelling of Earthquake Extremes	2
3.1	Choice of the POT-GPD Framework	2
3.2	Declustering	3
3.3	Threshold Selection	3
3.4	GPD Estimation	3
3.5	Model Diagnostics	5
3.6	Validation of Arrival Independence (Poisson Test)	5
4	Spatial Prediction	6
4.1	Temporal Layer: Hawkes Process Modelling	7
4.2	Spatial Layer: Geographic Location Modelling	7
4.3	Magnitude Layer: Conditional Magnitude Modelling	8
4.4	Three-Layer Joint Simulation Mechanism	8
4.5	Monte Carlo Forecasting Results	8
5	Expansion	11

1. Introduction

This project chooses an earthquake dataset as the object of study and applies Extreme Value Theory to analyze rare but high-impact major earthquake events. Earthquake data naturally exhibit characteristics of extreme events, making them one of the typical application areas of EVT. My objective is to use the statistical methods learned in the course to model the tail behavior of earthquake magnitudes and to provide a preliminary prediction of future extreme event risk.

Beyond the basic EVT modelling, I also further attempt to combine practical forecasting needs with simple model extensions and improvements. For example, in real-world applications, prediction results often concern not only the magnitude of an event, but also its time of occurrence and approximate spatial location. Therefore, in the later sections, I include richer modelling attempts in order to demonstrate the potential extension of EVT methods to practical problems.

2. Dataset and Preprocessing

2.1 Source and Scope

- Source: USGS Earthquake Catalog (<https://earthquake.usgs.gov/earthquakes/search/>)
- Region: Japan (30° – 46° N, 130° – 150° E)
- Time period: 2000.01.01–2026.02.26
- Minimum magnitude: 4.5

The reason for choosing the Japan region as the object of study is that seismic activity is frequent there and exhibits clear characteristics of extreme events. The tail sample is relatively rich, which makes the dataset suitable for extreme value analysis. In addition, the earthquake monitoring system in Japan and surrounding areas is well developed, and the quality of publicly available data is relatively high.

2.2 Selected Variables

The original dataset contains multiple earthquake observation variables. In this project, the following core fields are retained: occurrence time (`time`), latitude (`latitude`), longitude (`longitude`), depth (`depth`), magnitude (`mag`), and place description (`place`).

Among them, the magnitude variable is the core object of EVT modelling and is used for the subsequent threshold exceedance analysis and GPD fitting. The time variable is used to study the event arrival process and later time prediction. Latitude and longitude are used for spatial simulation in the extended modelling section. Depth and place description are mainly retained as auxiliary information for additional interpretation when necessary.

2.3 Cleaning Procedure

Before the formal modelling stage, preprocessing steps were carried out, including time format conversion, removal of missing values, and sorting observations by time. It should be noted that, since the sample was selected by a rectangular latitude–longitude boundary, some mistakenly included events from the Russian Far East region were mixed into the dataset. These observations were manually removed.

2.4 Descriptive Overview

After the cleaning procedure, the final sample contains 11,333 earthquake events. The average magnitude is approximately 4.82, the median is 4.70, and the maximum magnitude reaches 9.10, indicating that the dataset contains clear extreme major earthquake events.

From the magnitude distribution plot, it can be seen that most events are concentrated in the lower magnitude range, while the number of high-magnitude events decreases rapidly, showing a clear right-tail feature. This confirms the rare extreme-event structure of the dataset. The time scatter plot further shows that the Japan region remained highly seismically active throughout the study period, together with a small number of sudden high-magnitude events.

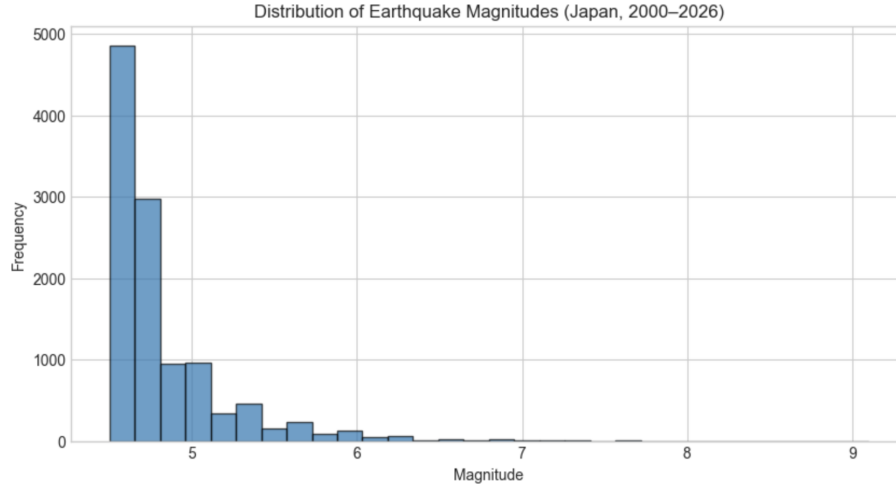


Figure 1: Distribution of Earthquake Magnitudes

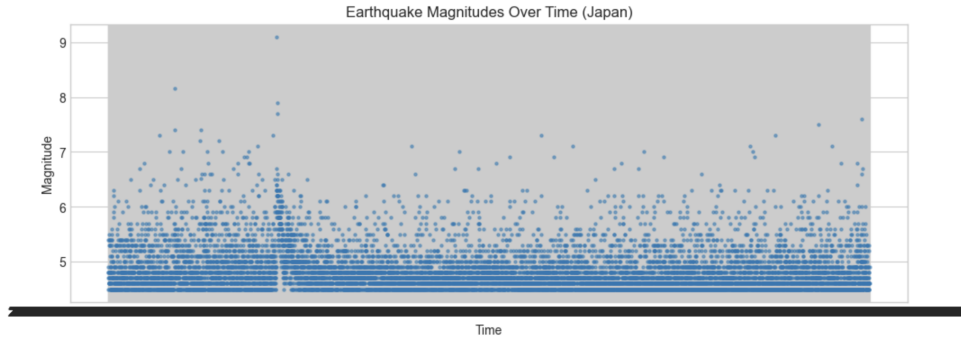


Figure 2: Earthquake Magnitudes Over Time

3. EVT Modelling of Earthquake Extremes

3.1 Choice of the POT-GPD Framework

This project adopts the POT-GPD approach, which consists of selecting a high threshold, retaining only events that exceed this threshold, and modelling the exceedances using the Generalized Pareto Distribution (GPD). The reason for choosing this method is that, for earthquake data, the main interest usually does not lie in the overall event distribution, but rather in the tail risk associated with a small number of high-magnitude events.

By contrast, the Block Maxima method uses only the maximum value within each time interval. If the time intervals are chosen too finely, many local maxima with low or moderate magnitudes may be included, which weakens the focus of extreme value analysis. If the intervals are chosen too broadly, the number of observations becomes limited. For data such as earthquakes, where extreme events are already relatively rare, this leads to inefficient use of available information. Therefore, the POT-GPD approach is more suitable for the characteristics of this dataset and the objective of this project.

However, it should be emphasized that the POT-GPD method is usually based on the assumption that exceedances above the threshold are approximately independent.

3.2 Declustering

Earthquake events usually exhibit clear clustering behaviour. For example, after a major earthquake occurs, it is often followed within a short period by multiple aftershocks or a sequence of related seismic events. Therefore, observations in the raw earthquake catalog are not mutually independent. If these observations are used directly for EVT modelling, tail samples may be repeatedly amplified, which can affect the stability of parameter estimation.

For earthquake data, declustering is a common practical compromise between the theoretical assumptions of EVT and the dependence structure observed in reality. Specifically, I apply a fixed time-window method: within each 24-hour time window, only the event with the largest magnitude is retained and treated as the representative event of that local seismic cluster.

After this step, the sample size is reduced from the original 11,333 observations to 3,776, indicating that a noticeable degree of event clustering exists in the raw catalog.

3.3 Threshold Selection

The choice of the threshold u is a central step in the POT-GPD framework and requires a balance between model bias and estimation variance. If the threshold is too low, the sample may not yet belong to the genuine tail region. If the threshold is too high, the number of exceedances available for estimation becomes too small, leading to unstable parameter estimates.

In this report, the threshold is assessed using three criteria. First, from the empirical quantiles, the 90%, 95%, and 99% quantiles are approximately $q_{0.90} = 5.30$, $q_{0.95} = 5.60$, and $q_{0.99} = 6.30$, which provide an initial reference range for candidate thresholds. Second, in the Mean Excess Plot, a relatively stable and approximately linear trend appears for $u \in [5.4, 5.8]$, suggesting that this interval is reasonably consistent with the GPD tail approximation. Finally, in the shape parameter stability plot, the shape parameter ξ fluctuates less over the interval $u \in [5.4, 5.7]$, and the confidence intervals remain relatively reasonable, indicating more stable parameter estimation.

Combining these results, the stable intersection region is identified as $[5.4, 5.7]$, and the final modelling threshold is chosen as $u = 5.5$.

In practice, I also tested the model with $u = 5.6$, and the results were broadly similar.

3.4 GPD Estimation

After selecting the threshold $u = 5.5$, all exceedances above this threshold were extracted, and the parameters of the Generalized Pareto Distribution (GPD) were estimated by Maximum

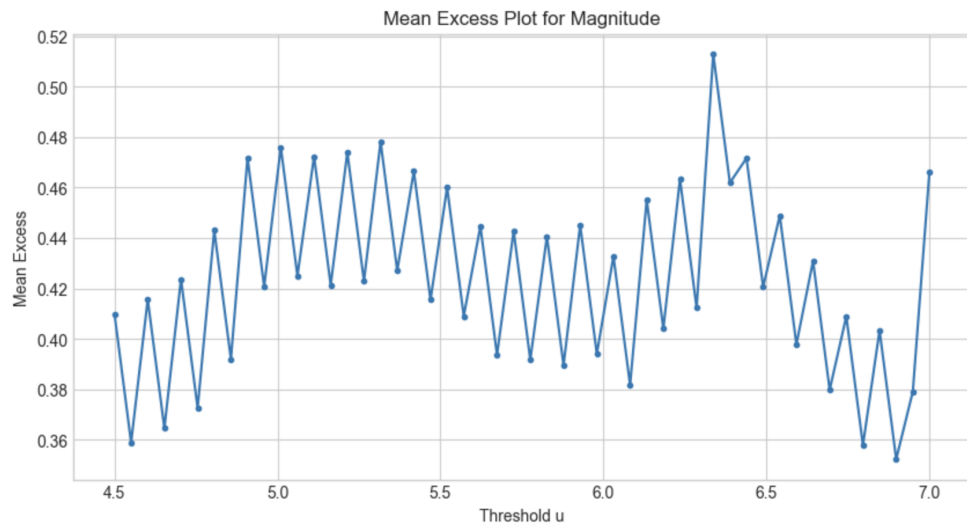


Figure 3: Mean Excess Plot for Magnitude

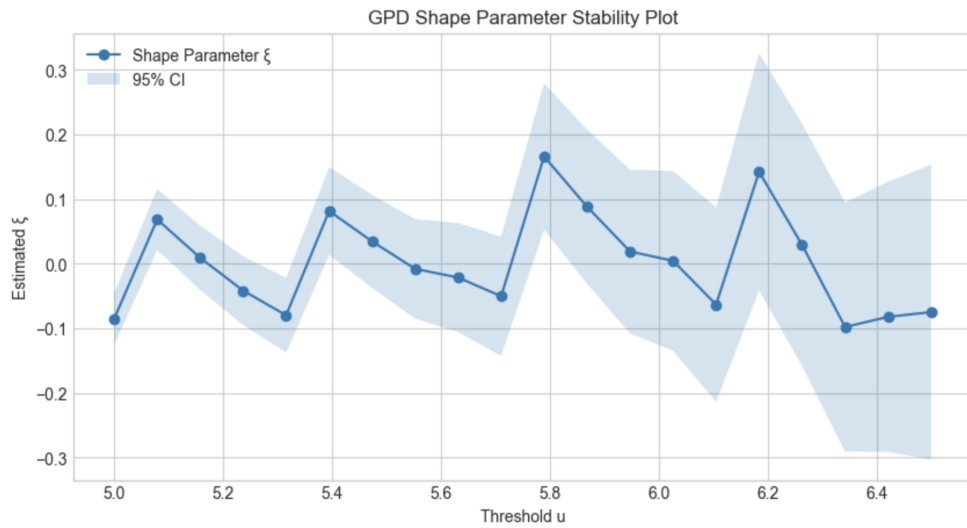


Figure 4: GPD Shape Parameter Stability Plot

Likelihood Estimation (MLE). The fitted results give a shape parameter and a scale parameter

$$\xi = -0.099, \sigma = 0.581$$

with a total of 413 exceedances.

The shape parameter ξ is slightly negative, indicating that tail growth is limited. Although extreme earthquake magnitudes still represent substantial risk, the tail does not display a strongly heavy-tailed behaviour. This result is reasonably consistent with the physical constraints that govern earthquake magnitudes.

Based on the estimated model, return levels were further calculated for different return periods. The estimated return levels for 10-year, 20-year, 50-year, and 100-year periods are approximately 7.81, 8.06, 8.37, 8.59, respectively.

3.5 Model Diagnostics

To evaluate the quality of the fitted model, the distribution of exceedances, together with a Q-Q plot and a P-P plot, were used as diagnostic tools for the GPD model.

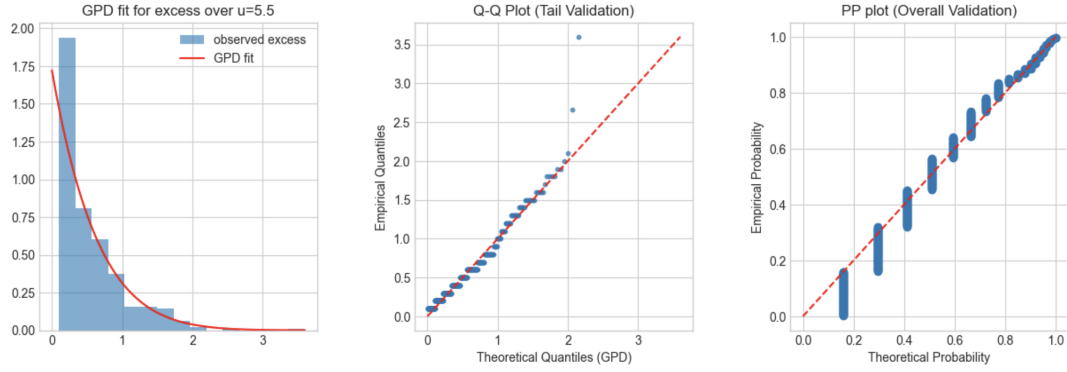


Figure 5: GPD Fitting Analysis

From the histogram of exceedances and the fitted density curve, the theoretical distribution matches the main body of the sample reasonably well, indicating that the model captures the overall shape of the exceedance data. In the Q-Q plot, most points lie close to the reference line, suggesting general agreement between theoretical and empirical quantiles. A certain deviation appears only in the highest tail region, which is common when the number of extreme observations is limited. The P-P plot also shows good consistency between empirical probabilities and theoretical probabilities, indicating that the overall model fit is relatively stable.

It should be noted that such tail deviations are common in extreme value analysis. They may arise from the limited number of extreme samples, threshold selection uncertainty, or remaining temporal dependence in the raw earthquake data, such as incompletely removed aftershock clustering effects. This also suggests that, although a static GPD model can describe the overall tail behaviour reasonably well, it still has limitations in capturing the temporal clustering structure of extreme events.

3.6 Validation of Arrival Independence (Poisson Test)

If exceedance events are approximately independent, their arrival times can usually be described by a Poisson process. In that case, inter-arrival times should be approximately exponentially

distributed, and the mean and variance of event counts over equal time intervals should be of similar magnitude.

In the previous diagnostic section, some deviation already appeared in the tail behaviour, and one possible explanation is the presence of remaining temporal dependence in the sample. We therefore examine this point more formally. According to the empirical results, the exponential Q-Q plot shows a reasonable fit in the central region, but a clear deviation in the tail, indicating that the observed inter-arrival distribution is heavier-tailed than the theoretical exponential distribution. In addition, the dispersion index computed both from inter-arrival based counts and from annual exceedance counts is significantly greater than 1, while for an ideal Poisson process it should be close to 1.

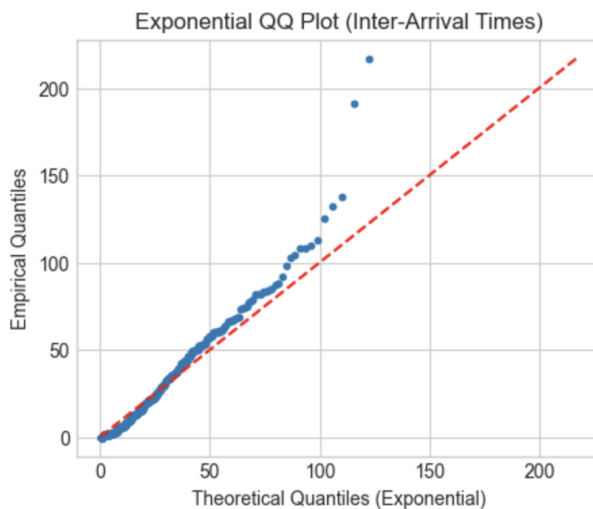


Figure 6: Exponential Q-Q Plot (Inter-Arrival Times)

These results suggest that the data still contain a certain degree of clustering, and the Poisson approximation is not fully satisfactory. This means that, although the POT-GPD model can describe the tail behaviour of earthquake magnitudes, its underlying independence assumption is not fully supported in the present dataset, so the estimation quality may be affected. Based on this observation, the next section introduces a spatio-temporal point process model capable of representing self-excitation and clustering effects, in order to build a forecasting framework that is closer to the real mechanism of earthquake activity.

4. Spatial Prediction

The POT-GPD method in the previous section mainly focuses on the tail distribution of earthquake magnitudes and is relatively effective in describing the intensity of extreme events, but it has limited ability to describe the occurrence time and spatial location of events. As shown in the previous section, earthquake data still exhibit a certain degree of clustering, so relying only on a static independence assumption is not fully sufficient.

Therefore, in this section, I further attempt to construct a more complete stochastic forecasting framework. In the time dimension, a Hawkes process is introduced to describe possible self-excitation and clustering features of earthquake events. In the spatial dimension, the latitude–longitude distribution of events is modelled to describe regions where future events may occur. In the magnitude dimension, the previous extreme value theory framework is retained to characterize the tail behaviour of earthquake magnitudes.

As discussed in the course, even when data do not satisfy approximate independence, extreme value theory may still be applied under certain conditions:

1. Regular variation: the tail distribution has a stable scaling structure in the asymptotic sense.
2. Mixing / weak dependence: observations far apart are approximately independent, without infinitely persistent strong dependence.
3. Anticlustering: extreme values do not cluster together without bound.

For earthquake data, these conditions are usually reasonable. The previous GPD Q-Q plot and P-P plot indicate that the tail of this dataset satisfies the regularity condition. After the Poisson validation, two further checks were added. The Hawkes process estimation satisfies the stability condition ($\alpha < \beta$), indicating that dependence mainly appears through aftershock effects following a main shock, which corresponds to short-term dependence. As time passes, the event intensity gradually declines and does not remain permanently at a high extreme level. At the same time, the spatial clustering test shows that large earthquakes have a relatively small average nearest-neighbour distance, indicating that extreme events are mainly concentrated near existing seismic belts rather than occurring randomly across space.

We now formally proceed to the modelling stage.

4.1 Temporal Layer: Hawkes Process Modelling

In the previous validation stage, clustering behaviour was examined in a preliminary way by plotting curves under fixed parameters. In this subsection, Maximum Likelihood Estimation is further applied to formally fit the Hawkes model, in order to quantify the temporal dependence structure.

The estimation uses 381 exceedance events from the training sample, corresponding to an observation period of approximately 8607 days. The mean inter-arrival time is about 22.65 days, with a standard deviation of 26.29 days, indicating clear variability in event arrivals.

The estimated parameters are

$$\hat{\mu} = 0.0242, \quad \hat{\alpha} = 0.0155, \quad \hat{\beta} = 0.0339.$$

These results confirm a moderate degree of temporal clustering for extreme earthquake events. Moreover, since $\hat{\alpha} < \hat{\beta}$, the model satisfies the stability condition, meaning that the effect of a single event gradually decays over time rather than accumulating indefinitely.

4.2 Spatial Layer: Geographic Location Modelling

In the spatial dimension, the latitude–longitude coordinates of exceedance events in the training set are modelled in order to describe regions where future extreme earthquakes are more likely to occur. This is closer to realistic earthquake forecasting scenarios.

As observed previously, the spatial structure of the data is clearly non-uniform. Therefore, a two-dimensional Kernel Density Estimation (KDE) is used to obtain a smooth estimate of the event location distribution, without assuming a simple uniform distribution.

The estimated spatial intensity surface shows that high-risk areas are mainly distributed in a belt along the trench zone east of Japan, especially offshore northeastern Honshu. This is broadly consistent with the actual seismic belt structure of Japan.

4.3 Magnitude Layer: Conditional Magnitude Modelling

I continue to use the GPD framework to model exceedance magnitudes. However, here the scale parameter is allowed to vary with time and spatial coordinates, in order to capture differences in extreme earthquake intensity across regions.

At this stage, the conditional structure only includes linear effects of longitude, latitude, and time. More complex factors such as variable interactions or Hawkes intensity effects are not yet included.

The parameters are estimated by Maximum Likelihood Estimation, and the model converges successfully. The estimated tail index is

$$\hat{\xi} = -0.1165.$$

This value remains negative, which is consistent with the conclusion of the previous static GPD model and suggests an upper bound in the magnitude tail.

The additional estimated coefficients are

$$b_0 = -0.6400, \quad b_t = -0.000026, \quad b_{\text{lon}} = -0.0103, \quad b_{\text{lat}} = 0.0445.$$

The spatial coefficients are relatively more pronounced, indicating that during the sample period, extreme earthquake intensity is more strongly influenced by geographic location. This is broadly consistent with the fact that seismic activity in Japan is mainly driven by tectonic plate structure. By contrast, the time coefficient is close to zero, suggesting that long-run temporal trends are not significant. This is again consistent with the previous evidence of mainly short-term dependence.

4.4 Three-Layer Joint Simulation Mechanism

After separately modelling the temporal layer, spatial layer, and magnitude layer, the three components are combined into a unified joint simulator for generating complete future paths of extreme earthquake events. As a result, each simulation produces a future event sequence including occurrence time, event location, and earthquake magnitude, thereby forming a more complete earthquake risk forecasting framework.

To evaluate predictive performance, the sample is divided into a training set (381 observations) and a test set (32 observations). The figure below presents the comparison between simulated and observed magnitude–time behaviour.

4.5 Monte Carlo Forecasting Results

To characterize the overall probabilistic behaviour of the forecasts rather than relying on a single simulated path, the joint simulator was repeated 300 times using Monte Carlo simulation.

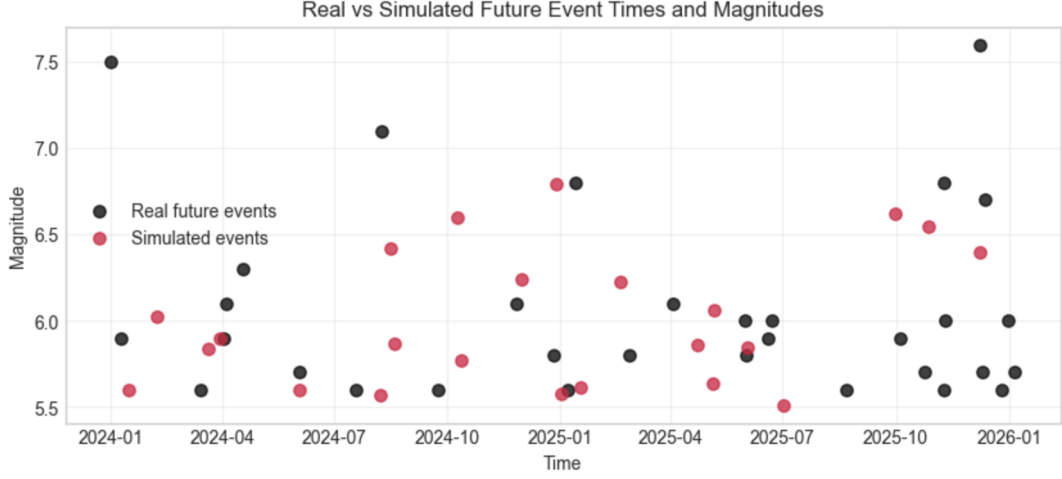


Figure 7: Real vs Simulated Future Event Times and Magnitudes

(1) Comparison of Observed Test Outcomes and Monte Carlo Forecast Distribution

Overall, the observed outcomes lie within a reasonable range of the simulated results and remain relatively close to the centre of the predictive distribution, suggesting that the joint model has reliable forecasting ability.

Metric	Observed	Mean	Median	5%	95%
Event count	32	26.47	25.00	14.00	44.00
Maximum magnitude	7.60	7.07	7.02	6.48	7.86
Mean magnitude	6.07	5.96	5.96	5.81	6.12
$P(\max M \geq 7)$	True	0.523	—	—	—

Table 1: Observed Test Outcomes vs Monte Carlo Forecast Summary

(2) Validation of Event Count Distribution

The histogram of simulated event counts shows that most outcomes are concentrated between 20 and 35 events, while the observed number of test events is 32 (red dashed line). This indicates that the real outcome lies in a high-probability region of the model, suggesting that the forecast of total future event count is reasonable, without clear overestimation or underestimation.

(3) Validation of Maximum Magnitude Distribution

The centre of the simulated maximum magnitude distribution lies approximately between 6.8 and 7.3, while the observed maximum magnitude in the test set is 7.60 (red dashed line). The observed value lies on the right side of the simulated distribution but remains within a reasonable range, indicating that the model is able to capture the possibility of strong extreme events while preserving tail risk uncertainty.

(4) Validation of Spatial Hotspot Map

The comparison between the average forecast hotspot map and the observed locations of test events shows that the high-risk areas identified by the model mainly follow the major seismic

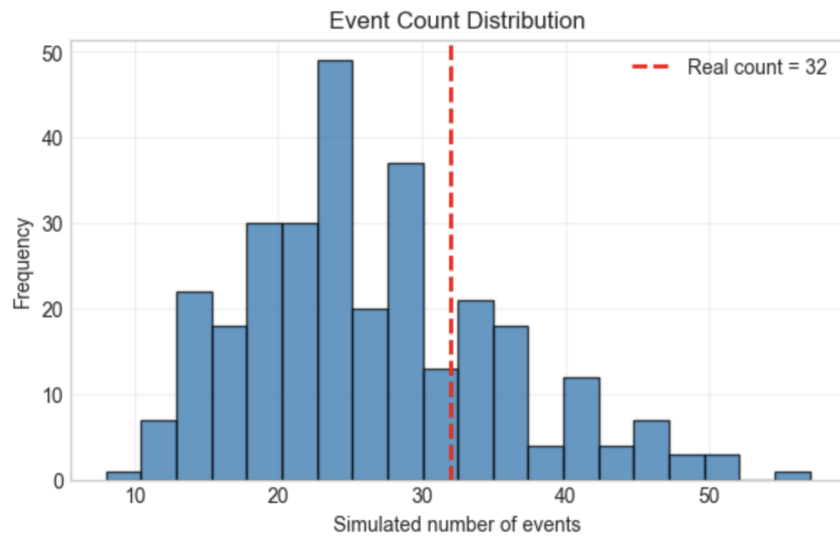


Figure 8: Distribution of Simulated Event Counts

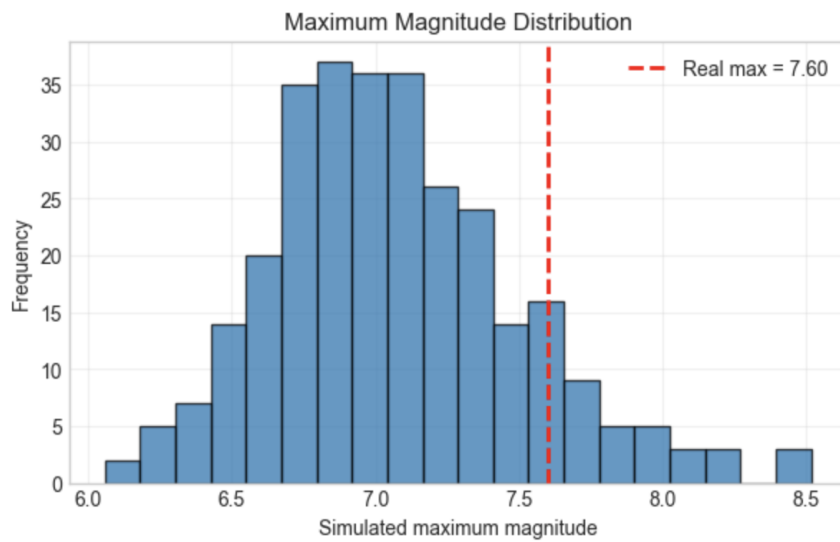


Figure 9: Distribution of Simulated Maximum Magnitudes

belts of Japan, especially offshore eastern Honshu. This is highly consistent with the spatial distribution of the observed events, indicating that the model successfully captures the spatial structure of the data.

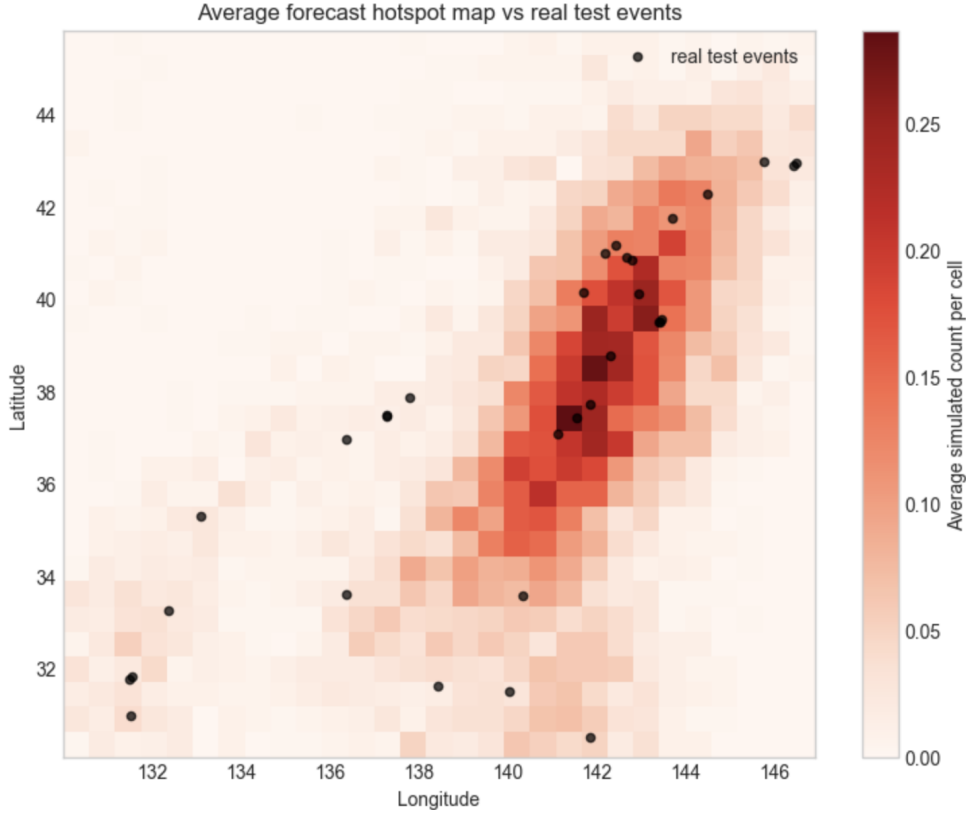


Figure 10: Average Forecast Hotspot Map vs Observed Test Events

Based on the combined visualisation results, the predictive performance of this three-layer joint model is significantly improved compared with the static GPD model. It is able to incorporate the three dimensions of event occurrence time, event location, and earthquake magnitude into a unified forecasting framework in a relatively complete manner, while achieving reasonably satisfactory simulation results.

At the same time, some limitations can still be observed in the forecasts. For example, the predicted average event intensity is slightly lower than the real level, the tail behaviour of extreme values remains relatively conservative, and some highly extreme scenarios are still insufficiently captured. This does not imply a failure of the simulation method itself, but is more likely due to the moderate simplifications made in the current modelling design regarding the number of covariates, the conditional distribution structure, and the complex interactions between variables. Overall, the model already provides an interpretable statistical framework for earthquake risk forecasting with substantial room for further extension.

5. Expansion

Although the final three-layer forecasting framework is already able to jointly describe the temporal, spatial, and magnitude characteristics of earthquake events, it still relies on a relatively simplified structure. For forecasting complex data such as earthquakes, many directions for further improvement remain possible.

- **Non-stationary Hawkes Modelling:** This report uses a stationary Hawkes process with fixed parameters. Future work could allow the background intensity or self-excitation parameters to vary over time, in order to capture changes in seismic activity across different periods and possible long-run cycles.
- **More Complete Joint Modelling:** The current framework uses a layered structure for time, space, and magnitude, which has the advantages of simple implementation and strong interpretability. Future work could develop a fully unified joint model in which these three components interact with one another, while incorporating additional relevant variables for a richer global representation closer to the real earthquake mechanism.
- **More Flexible Conditional GPD Modelling:** This report only uses a linear structure with a limited number of covariates. Future work could include interaction terms, non-linear effects, or additional covariates such as focal depth and Hawkes intensity, in order to improve the modelling of differences in extreme magnitudes and enhance forecasting accuracy.